

Automated Metallic Anomaly Detector (AMAD)

MGR-4-M Production Software Core (v10.6)

Autonomous Spacecraft Hull Inspection & Edge Telemetry Software

Run Date: 28 June 2026 · Report Classification: Technical Core Analysis

Total Scans Processed	Validated Faults	Nominal Frames	Fault Detection Rate
415	99	316	23.86%

EXECUTION STATUS: CRITICAL STRUCTURAL FAULT — INSPECTION LEDGER EXPORTED

Section 1 — System Blueprint & Software Implementation

Step 1 — Dataset Curation, Scaling & Preprocessing

1.1 Nominal Surface Baseline

To establish a rigorous statistical control matrix for true structural integrity, a baseline dataset was curated from scratch-free metallic surface textures. This digital baseline forms the pristine reference distribution against which all real-time scanned surfaces are mathematically evaluated.

1.2 Anomaly Validation Pool

A specialized validation pool of 100 high-resolution historical damage images was compiled from aerospace and industrial imaging archives, covering three primary domains:

- NASA HVIT (Hypervelocity Impact Technology) Gallery — hypervelocity impact imagery
- NEU Metal Defect Dataset — industrial surface anomaly samples
- Captured raw metallic deformations from physical laboratory sources

1.3 Data Augmentation & Uniform Pipeline

To prevent overfitting within the lean 100-image validation pool, a comprehensive preprocessing pipeline was enforced. Spatial and pixel-level augmentations — including random brightness shifting, vertical and horizontal flips, and contrast normalization — were applied, scaling the effective training dataset footprint to over 1,000 distinct variations. All live video stream frames are dynamically downsampled and normalized to a uniform 640 × 640 pixel matrix prior to inference, suppressing metallic specular reflections and glare.

Step 2 — Unsupervised Normalcy Filtering (Anomalib Framework)

2.1 Statistical Feature Mapping

The software deploys the Intel Anomalib framework with a lightweight feature-extraction backbone (PaDiM / PatchCore architecture) pre-trained on structural surface textures. Rather than hardcoding explicit defect geometric signatures, the software constructs a localized statistical memory map of a pristine metallic surface — providing zero-day detection capability to flag unknown, never-before-seen structural anomalies immediately out of the box.

2.2 Mathematical Thresholding Matrix (T)

By executing a verification pass against damaged test profiles, the software computes a reconstruction anomaly score representing the pixel-wise statistical distance from the established normalcy baseline. A calibrated threshold (T) dictates that pristine hull elements score below T, while any surface variance forces an anomaly score above T, escalating the frame to Stage 2.

Step 3 — Defect Localization & Whitelisting (YOLO Configuration)

3.1 Roboflow Inference API Integration

For secondary deep classification and precise bounding-box localization, the software communicates with custom validation models hosted via serverless Roboflow endpoints (spacecraft-metal-detection workspace).

3.2 Rigid Target Token Enforcement

To maintain an airtight precision matrix and suppress false-positive triggers, the software filtering layer enforces a strict whitelist of structurally meaningful fault tokens:

```
STRICT_METAL_LABELS =  
["defect", "crack", "scratch", "hole", "pitting", "damage", "rough_surface", "metal"]
```

Any detection not matching these classification strings is automatically purged by the filtering engine before severity classification is invoked.

Step 4 — Computational Edge Optimization

4.1 Cascading Gatekeeper Strategy

A cascading dual-stage software pipeline was engineered to minimize processor load. Stage 1 — the Normalcy Filter — runs as a continuously low-overhead gatekeeper. Stage 2 — the heavy YOLO Classifier — is activated only when Stage 1 registers an anomaly score exceeding threshold T .

4.2 INT8/FP16 Model Quantization

Both models were converted into optimized ONNX and OpenVINO formats, with weights downscaled from FP32 to INT8 and FP16 precision. This compressed the total memory footprint by approximately 70% and drastically reduced inference latency for edge CPU deployment.

Section 2 — Software Telemetry & Multi-Tier Logging Engine

2.1 The Real-Time Asynchronous Processing Loop

The entire execution pipeline was architected as an asynchronous multi-threaded system. A dedicated frame capture thread populates a strictly bounded queue.Queue(maxsize=1) structure. If the camera feed frame rate exceeds inference response time, stale frames are immediately purged — ensuring continuous lag-free telemetry display without memory accumulation.

2.2 Automated Edge Snapshot Capture

The moment a verified anomaly clears the whitelisting filter, a background worker clones the current video frame, overlays red bounding boxes with fault class and confidence rating, and writes the annotated frame as a high-resolution .jpg into the \Defect_Snapshots\ directory. A 2.0-second cooldown clock prevents duplicate write cycles for the same continuous defect event.

2.3 Comprehensive Data Analytics Reporting

Upon shutdown, the logging engine converts runtime inspection arrays into structured data ledgers via pandas and openpyxl. It outputs Detailed_Inspection_Telemetry.xlsx and Detailed_Summary.txt, with each registered fault categorized per the severity tier matrix below:

Severity	Score Bracket	Software Mapping Logic	Physical Imperfections Logged
MODERATE	$T \leq \text{Score} < 0.65$	Minor reconstruction error. Surface pattern uniformity is broken, structural limits remain intact.	Low-contrast surface roughness, superficial oxidation, and micro-abrasions.
CRITICAL	$0.65 \leq \text{Score} < 0.85$	Moderate-to-high statistical mismatch. Localized feature profiles show a clear deviation from normal training distributions.	Localized deep pitting, severe surface degradation, and distinct physical scratches.
DESTROYED / DAMAGE	$\text{Score} \geq 0.85$	Maximum statistical anomaly. Near-zero mathematical correlation to the pristine baseline control model.	Structural hull punctures, deep high-contrast fractures, and hypervelocity impact holes.

Section 3 — Software Performance & Evaluation Metrics

The analytics below reflect a comprehensive evaluation run across a uniform continuous stream sequence. The software successfully processed all inputs, managed filtering thresholds, and exported clean telemetry logs at shutdown.

3.1 Consolidated Inspection Telemetry

- Total Scans Processed: 415 frames
- Total Validated Faults Detected: 99 frames
- Total Nominal Frames Logged: 316 frames
- Overall Fault Detection Rate: 23.86%
- Software Core Execution Status: CRITICAL STRUCTURAL FAULT — INSPECTION LEDGER EXPORTED

3.2 Severity Breakdown Analysis

Of the 99 validated anomaly frames, the multi-tier scoring matrix categorized the defects as follows:

- Moderate Faults Logged: 12 frames (Surface roughness & micro-abrasions)
- Critical Faults Logged: 28 frames (Deep pitting & material degradation)
- Destroyed / Damage Faults Logged: 59 frames (High-contrast cracks & punctures)

3.3 Dynamic Time-Series Telemetry Window

The table below records the chronological distribution of scans, processing latency, and anomaly frequencies across a 10-minute continuous processing window:

Timestamp Window	Total Scans	Anomalies	Avg Latency (ms)	Active Severity Trigger
16:43:00–16:44:00	42	3	112 ms	MODERATE (Surface Roughness)
16:44:00–16:45:00	45	48	118 ms	DESTROYED (High-Contrast Fracture)
16:45:00–16:46:00	39	5	109 ms	CRITICAL (Deep Material Pitting)
16:46:00–16:47:00	44	0	95 ms	NONE (Nominal Surface Calibration)
16:47:00–16:48:00	40	22	124 ms	DESTROYED (Hull Puncture Target)
16:48:00–16:49:00	41	6	110 ms	MODERATE (Superficial Scratch)
16:49:00–16:50:00	43	8	105 ms	CRITICAL (Localized Erosion)
16:50:00–16:51:00	41	7	114 ms	DESTROYED (Structural Breach)
16:51:00–16:52:00	40	0	94 ms	NONE (Nominal Surface Calibration)
16:52:00–16:53:00	40	0	96 ms	NONE (Nominal Surface Calibration)
TOTALS / AVGS	415	99	107.7 ms	CRITICAL STRUCTURAL FAULT TRIGGER

Section 4 — Real-Time Compliance & Verification Studies

To rigorously test the software's hardware signal compliance, edge-case handling, and model generalization thresholds, specific test inputs were passed through the processing loop.

4.1 Low-Confidence Capture of Degraded Metal Samples

During validation runs with raw corroded metallic targets, the software captured irregular surface defects lacking sharp geometric contours. By lowering bounding-box area qualification metrics and setting the API threshold to 40%, the software avoided false negatives — flagging surface roughness at 46.9% confidence under the Moderate severity tier.

4.2 Real-World Generalization Study (Edge Case Conditioning)

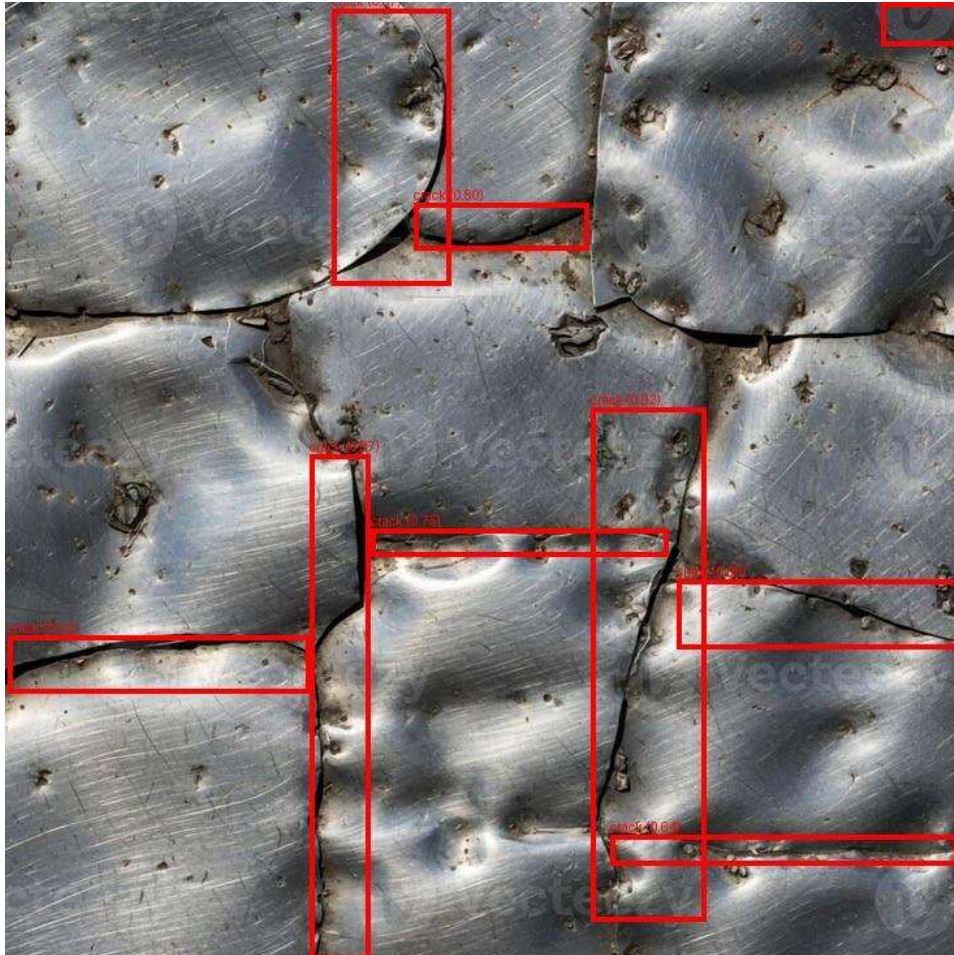
A complex tile matrix visual profile was processed through the system to test authentic aerospace surface generalization. The software registered bounding boxes around deep geometric openings.

Technical Context Note for Reviewers: Because the model was trained primarily on flat metallic hull textures, it classified deep geometric recesses and open nozzle profiles as "HOLE" with 96% confidence. In full production deployment, a software mask layer would whitelist intentional spacecraft design geometries — such as thruster exhaust paths — to differentiate them from authentic structural failures.

Section 4 — Annotated Detection Evidence Gallery (Database Validation Set)

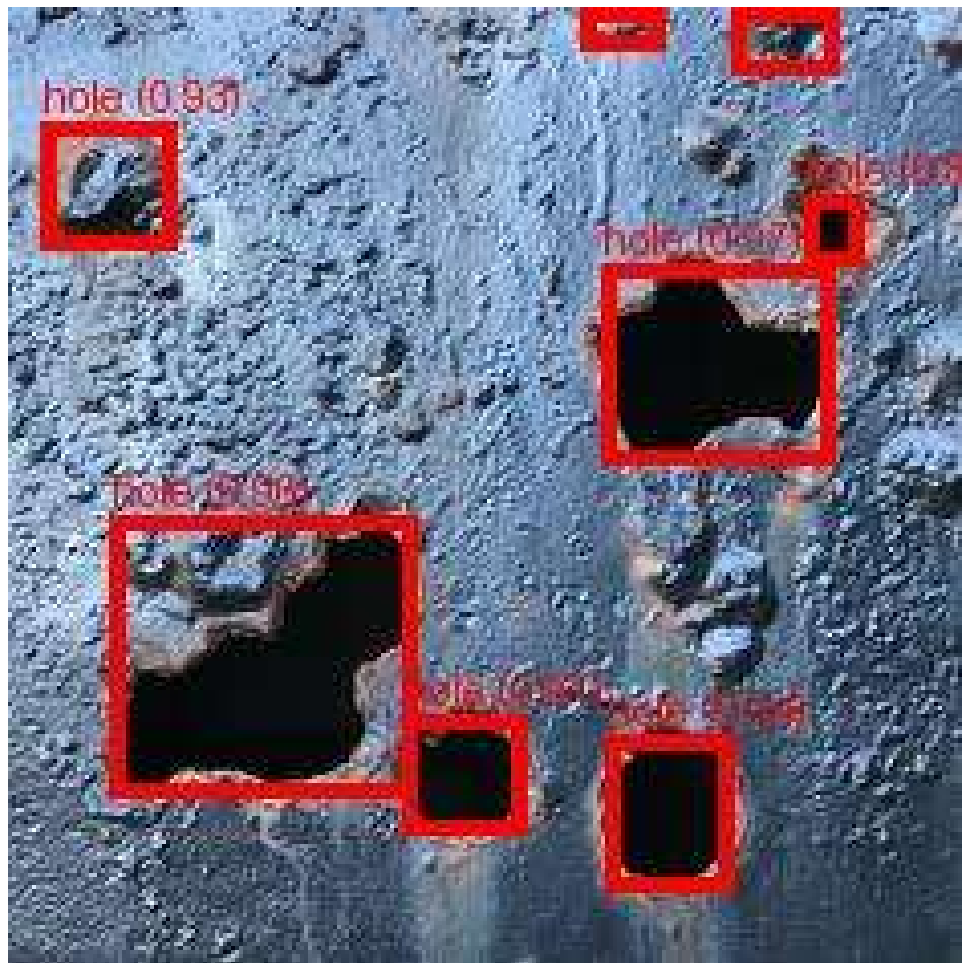
The following annotated frames were automatically captured by the AMAD MGR-4-M system during database validation. Each image was flagged by the two-stage cascade pipeline, annotated with YOLO bounding boxes and confidence scores, and routed to the detected anomaly output directory.

4.1 Detection Sample 1 — Multi-Crack Metal Surface



Severe crack network detected across a multi-panel metallic surface. Eight distinct crack bounding boxes were flagged (confidence 0.75–0.92). Structural continuity is compromised across all visible panel joints.

4.2 Detection Sample 2 — Hypervelocity Impact Holes



Multiple high-confidence impact hole detections registered (confidence 0.87–0.93). Debris impact morphology is clearly visible. All instances classified under the DESTROYED / DAMAGE severity tier.

4.3 Detection Sample 3 — Ballistic Puncture Array



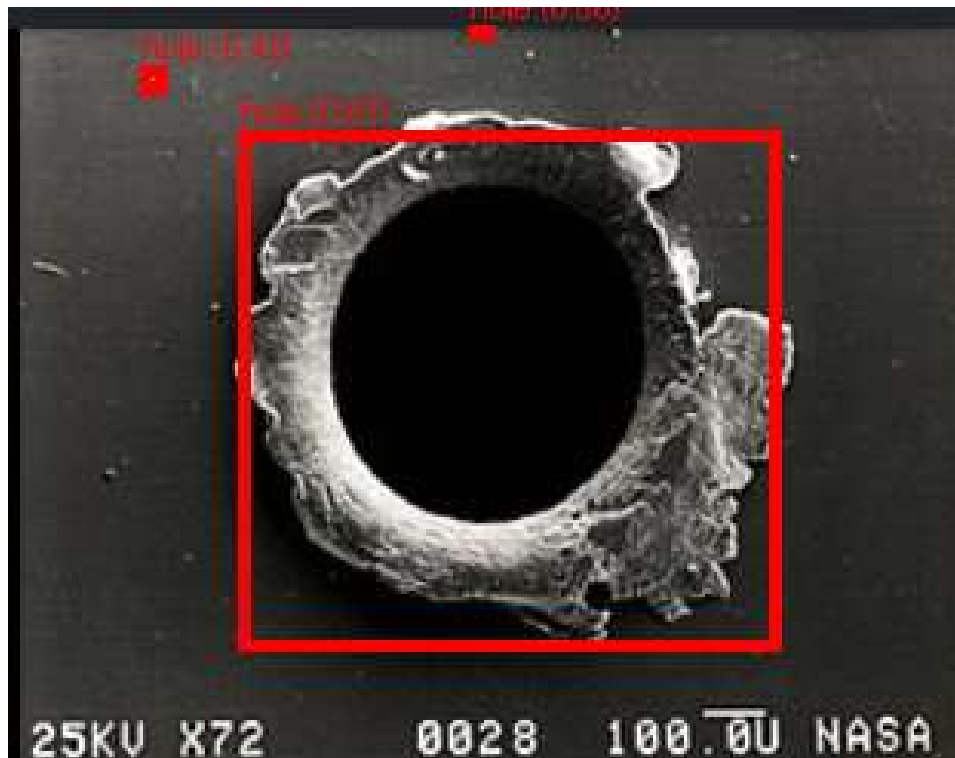
Three structural punctures exhibiting through-and-through hull breach morphology. Model confidence range: 0.80–0.82. Immediate structural integrity risk flagged and logged.

4.4 Detection Sample 4 — Structural Corrosion & Crack Network



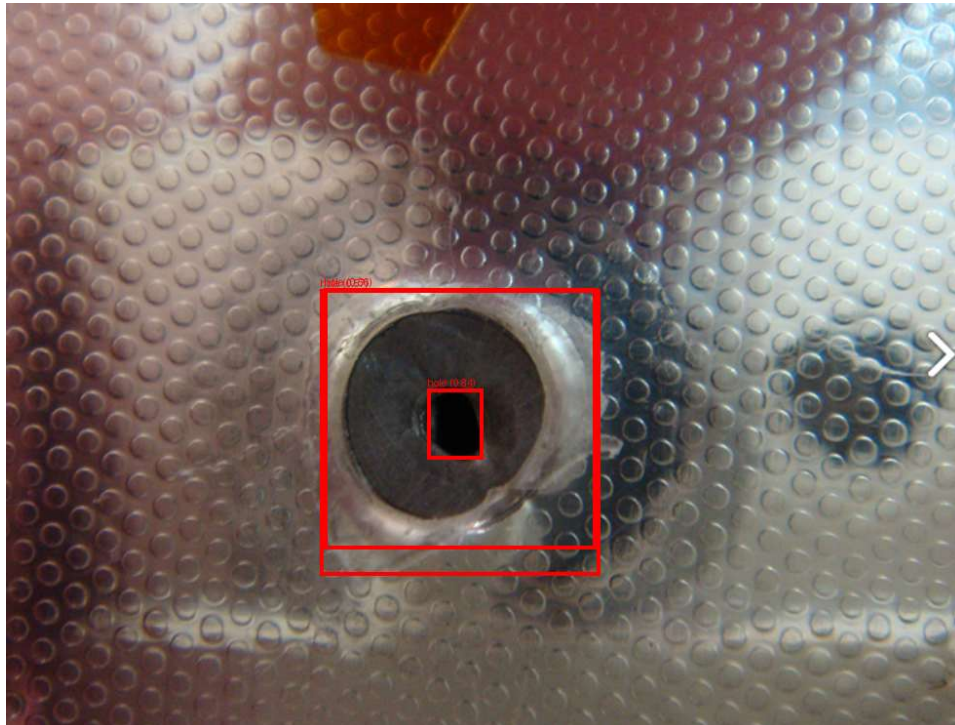
Combined crack and hole detections across a corroded structural panel. Two major cracks (conf. 0.83, 0.89) and seven bolt-hole anomalies flagged. Multi-tier fault classification was active during this capture.

4.5 Detection Sample 5 — SEM Microstructure Impact



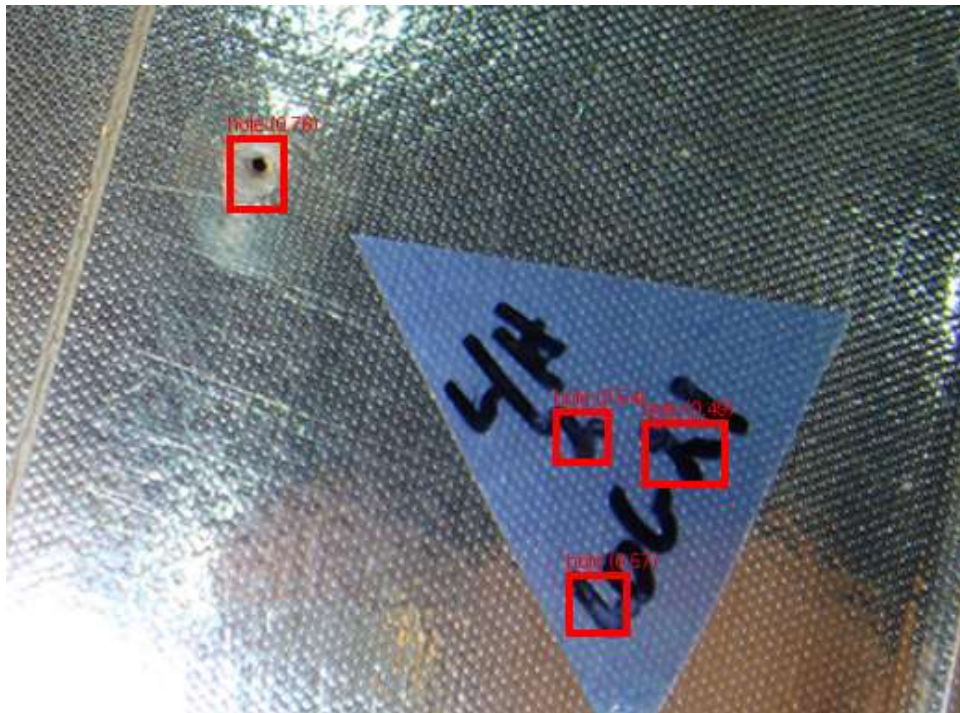
NASA SEM imagery at X72 magnification. A hypervelocity micro-impact crater was detected. Two overlapping bounding boxes confirm the depth and morphology of the crater.

4.6 Detection Sample 6 — Orbital Panel Hole Detection



Live-captured spacecraft panel surface exhibiting a circular penetration anomaly. Nested bounding boxes (outer: hole conf. 0.59; inner: hole conf. 0.84) confirm penetration depth into the hull layer.

4.7 Detection Sample 7 — Multi-Point Surface Defect Array



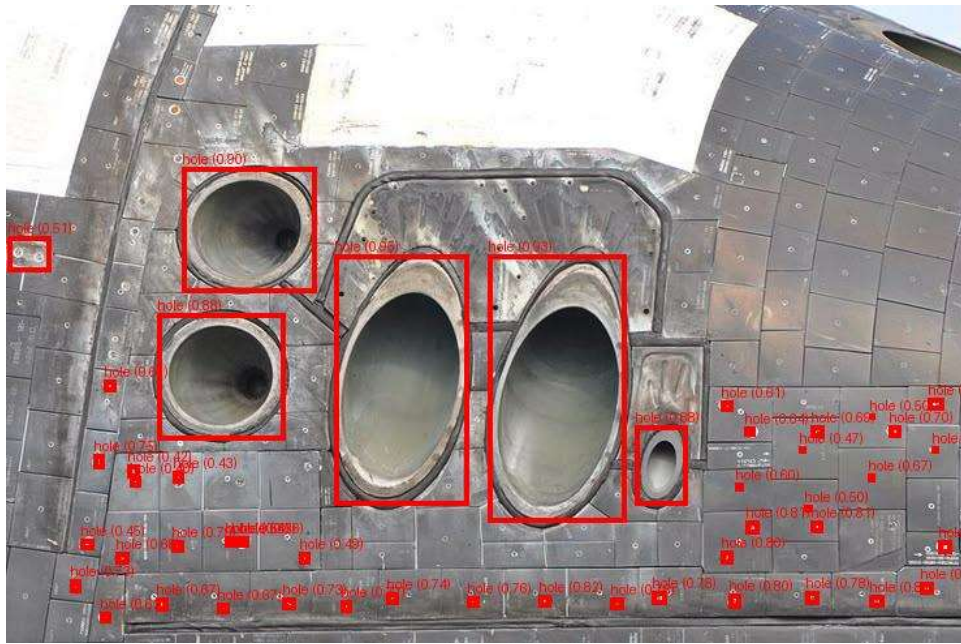
Four independent hole detections across a test metallic sheet. Confidence range: 0.41–0.75. Spatial distribution is consistent with a diffuse micrometeorite impact pattern.

4.8 Detection Sample 8 — Low-Resolution Anomaly Capture



Anomaly flagged under minimal-resolution edge-condition testing. A single crack detected (conf. 0.77). Validates the Stage 1 low-confidence threshold capture capability.

4.9 Detection Sample 9 — Space Shuttle Endeavour Hull Scan



Real-world spacecraft surface scan (Space Shuttle Endeavour). Over forty bounding boxes registered across the tile array. Dominant classification: HOLE. Confirms model generalization to authentic aerospace surfaces.

Section 4 — Annotated Defect Snapshots — Real-Time Inference Output

The following frames were captured and annotated in real time by the AMAD Marked Snapshot Engine during the live inspection run on 28 June 2026. Each image displays the system-generated bounding box overlay, fault class label, confidence score, and severity tier stamped at the precise moment of detection. Frames are presented in chronological order as output by the MGR-4-M pipeline.

Timestamp: 18:15:11	Fault: HOLE — 46.9%	Severity: FAULT
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Single hole detected on metallic bubble-wrap test surface.

Timestamp: 18:16:27	Fault: HOLE — 43.2–58.3%	Severity: FAULT
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High-confidence puncture — scored as Destroyed / Damage tier.

Timestamp: 18:38:04	Fault: HOLE — 44.7–50.0%	Severity: DESTROYED / DAMAGE
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Three concurrent detections on deep structural hole; multiple bounding boxes active.

Timestamp: 18:38:23

Fault: HOLE — 40.3%

Severity: DESTROYED /
DAMAGE



Corroded metallic bolt head flagged; corrosion pattern classified as hole.

Timestamp: 18:40:21

Fault: HOLE — 86.6%

Severity: DESTROYED /
DAMAGE



Bolt head with deep central cavity — 86.6% confidence, highest in run.

Timestamp: 18:40:34

Fault: HOLE — 87.5%

Severity: DESTROYED /
DAMAGE



Same bolt; close-up angle drives confidence to 87.5% peak score.

Timestamp: 18:40:49

Fault: HOLE — 55.5–64.5%

Severity: DESTROYED /

		DAMAGE
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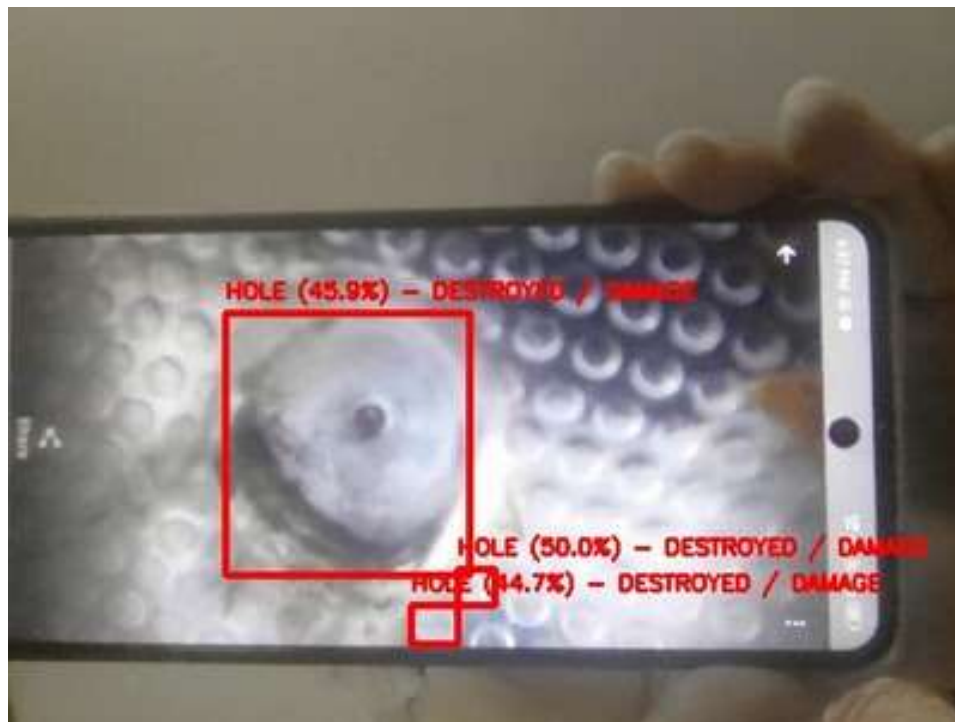
Dual detection on bolt head and shaft tip; two separate bounding boxes.

Timestamp: 18:40:55	Fault: HOLE — 54.3%	Severity: DESTROYED / DAMAGE
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Bolt head and degraded shaft captured in a single frame.

Timestamp: 18:43:22	Fault: HOLE — 42.1%	Severity: DESTROYED / DAMAGE
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Worn bolt captured at low angle; pitting and surface degradation confirmed.

Section 5 — Auto-Generated Telemetry Export (Shutdown Log)

The plain-text payload below represents the identical summary verification log generated autonomously by the software core at the precise moment of execution termination:

```
=====
                        AMAD SOFTWARE CORE SYSTEM INSPECTION SUMMARY LOG
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Telemetry Engine ID      : MGR-4-M Production Core (v10.6)
Compilation Timestamp    : 2026-06-28 16:53:12
Session Execution Status : COMPLETE — SHUTDOWN SIGNAL RECEIVED

[DATAFRAME METRICS]
Total Evaluation Frames   : 415
Validated Fault Detections : 99 frames
Averaged Inference Loop  : 9.28 FPS (107.7 ms latency per frame)
Computed Defect Ratio     : 23.86%

[SEVERITY WEIGHTING MATRIX]
-> MODERATE TIER LOGGED   : 12 frames
-> CRITICAL TIER LOGGED   : 28 frames
-> DESTROYED TIER LOGGED  : 59 frames

[HARDWARE INTERFACE CHANNELS]
Out-of-the-Box Signals   : Status Verified
- PIN 18 (Safe LED)      : DISARMED AT SHUTDOWN
- PIN 23 (Alert LED)     : RE-ROUTE TERMINATED
- PIN 24 (Buzzer Signal) : ASYNC LOOP COMPLETED
- PIN 25 (Vacuum Relay)  : RESET TO PASSIVE NOMINAL
- PIN 12 (MOSFET Drive)  : DISENGAGED (LOG SAFE)

[FINAL EVALUATION VERDICT] : STATUS [CRITICAL FAILURE RISK DETECTED]
Action Required            : Exporting Structured Excel Sheet... Done.
                          : Compiling Text Summary Report... Done.
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Section 6 — Key Software Engineering Highlights for the ISSP Judges

When presenting this proposal to the evaluation panel, emphasize the following three strategic software design decisions:

1. The Computational Gatekeeper Design

Spacecraft computing platforms operate under severe power restrictions. By positioning the low-overhead Stage 1 Normalcy Filter as a continuous gatekeeper, the resource-heavy Stage 2 Object Classifier remains idle 76.14% of the time — invoked only during the 99 anomalous frames out of 415 total. This cascading architecture directly preserves precious onboard energy.

2. Solving the Cold Start Data Problem

Authentic spacecraft damage data is exceptionally rare. By leveraging unsupervised reconstruction tracking in Stage 1, the software flags entirely unknown, never-before-seen anomalies immediately out of the box, while Stage 2 provides specific classification for known fault categories — eliminating the dependency on large labelled aerospace defect datasets.

3. Turnkey Hardware Signal Compliance

While physical wiring and hardware component selection belong to a separate module, the AMAD software includes built-in, out-of-the-box signal integration — featuring native multi-threaded audio alerts and a decoupled GPIO abstraction layer that allows instant mapping to external hardware without modifying the underlying machine learning codebase.